

Middleware Architectures 1

Lecture 6: Communication Protocols

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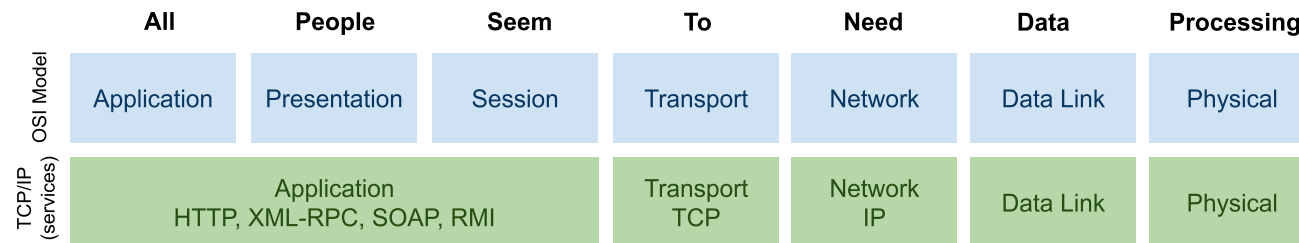
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Overview

- Introduction to Application Protocols
- Hypertext Transfer Protocol (HTTP)
- Security

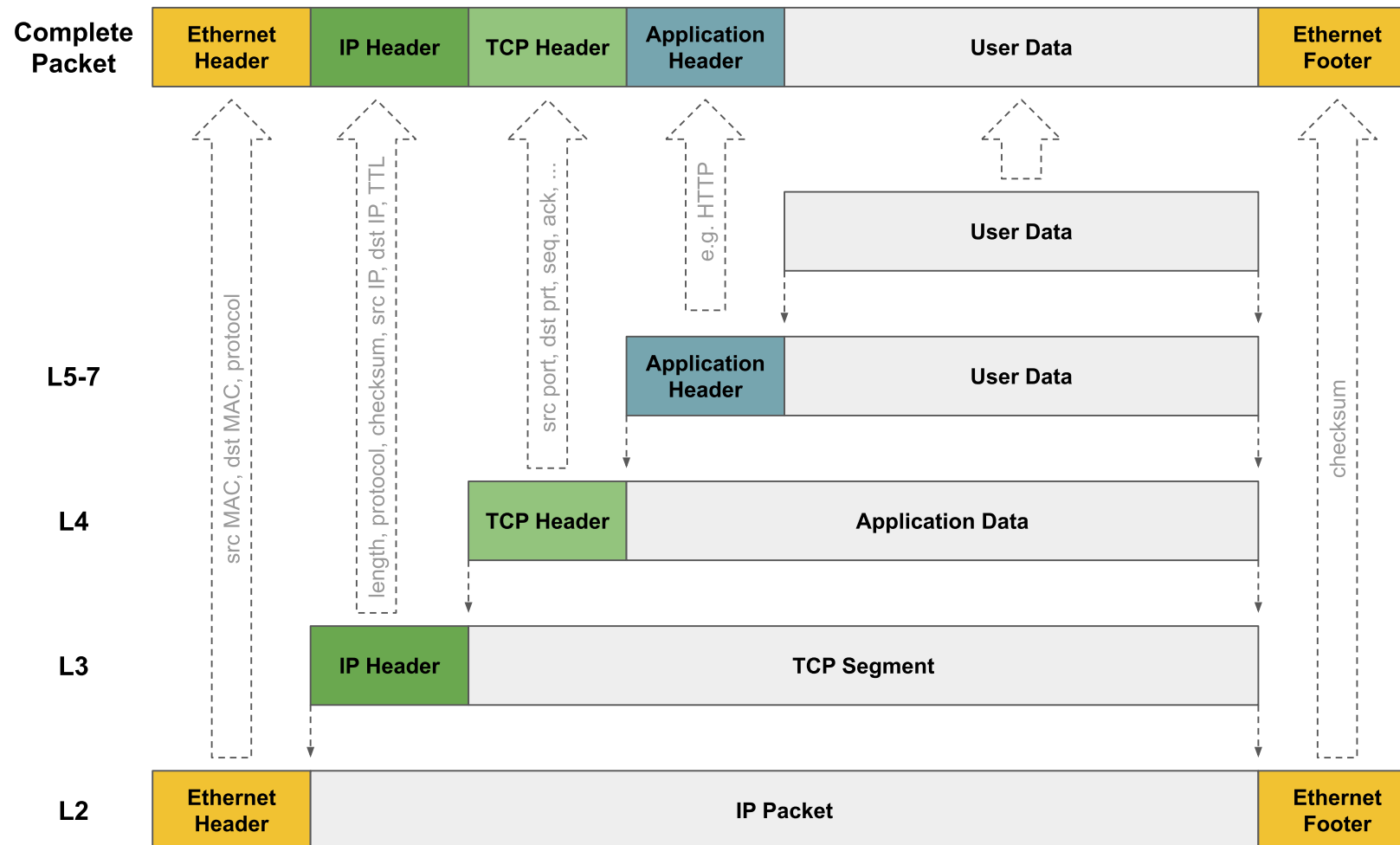
Application Protocols

- Remember this

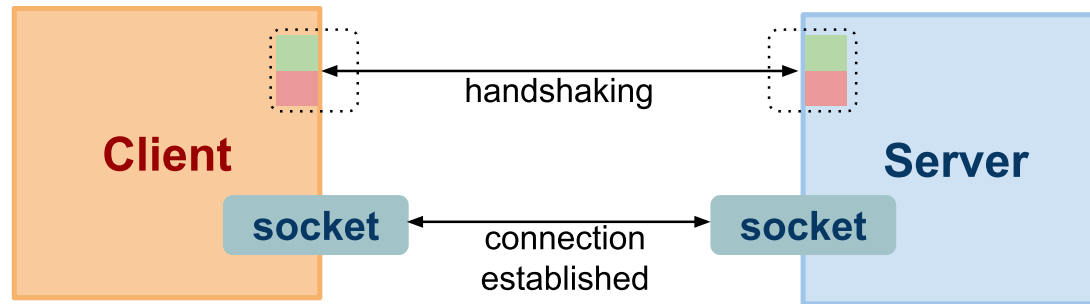


- App protocols mostly on top of the TCP Layer
 - *use TCP socket for communication*
- Major protocols
 - *HTTP – most of the app protocols layered on HTTP*
 - *HTTP/2 – new binary framing*
 - *gRPC – modern RPC protocol on top of HTTP/2*
 - *WebSocket – new protocol part of HTML5*
- Legacy
 - *RMI – Remote Method Invocation*
 - *Java-specific; vendor-interoperability problem*
 - *XML-RPC and SOAP – Remote Procedure Call and SOAP*

Anatomy of a Packet



Socket



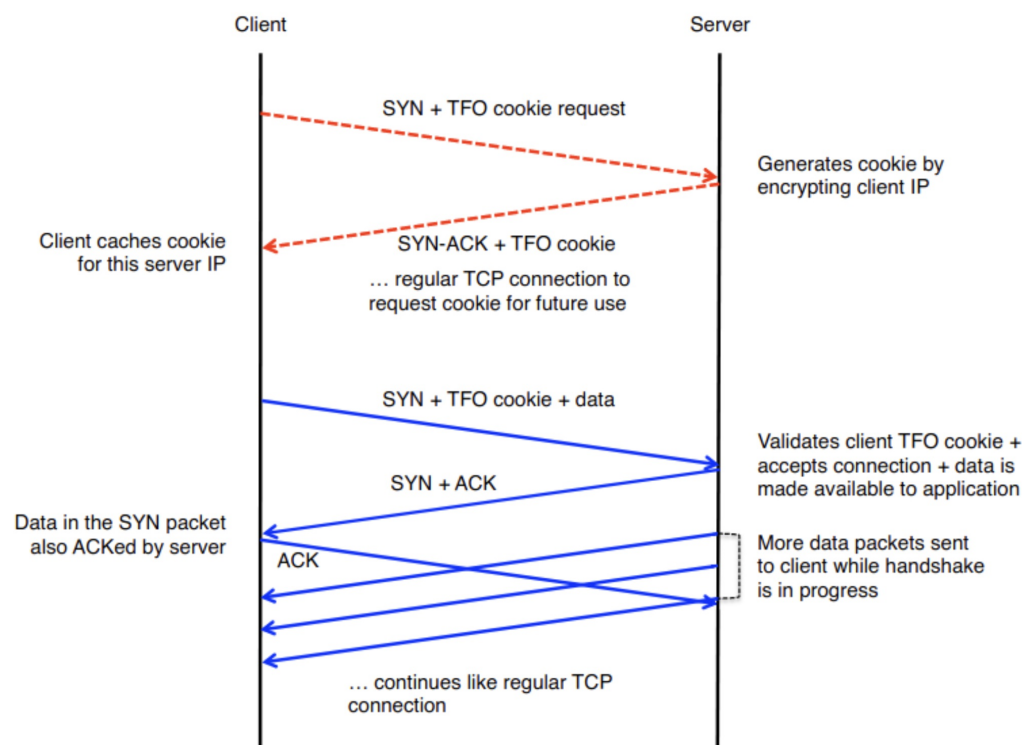
- Handshaking (connection establishment)
 - The server listens at `[dst_ip, dsp_port]`
 - Three-way handshake:
 - the client sends a connection request with TCP flags (SYN, $x=rand$)
 - the server responds with its own TCP flags (SYN ACK, $x+1$ $y=rand$)
 - the client acknowledges the response, can send data along (ACK, $y+1$ $x+1$)
 - Result is a socket (virtual communication channel) with unique identification:
`socket=[src_ip,src_port;dst_ip,dst_port]`
- Data transfer (resource usage)
 - Client/server writes/reads data to/from the socket
 - TCP features: reliable delivery, correct order of packets, flow control
- Connection close

New Connection Costs

- Creating a new TCP connection is expensive
 - *It requires to complete a full roundtrip*
 - *It is limited by a network latency, not bandwidth*
- Example
 - *Distance from London to New York is approx. 5500 km*
 - *Communication over a fibre link will take at least 28ms one way*
 - *Three-way handshake will take a minimum of 56ms*
- Connection reuse is critical for any app running over TCP
 - *HTTP Keep-alive*
 - *HTTP pipelining*

TCP Fast Open

- TFO allows to speed up the opening of successive TCP connections
 - *TCP cookie stored on the client that was established on initial connection*
 - *The client sends the TCP cookie with SYN packet*
 - *The server verifies the TCP cookie and can send the data without final ACK*
 - *Can reduce network transaction latency by 15%*



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- **Hypertext Transfer Protocol (HTTP)**
 - *Performance*
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 - *HTTP in Kubernetes*
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Hypertext Transfer Protocol – HTTP

- Application protocol, basis of Web architecture
 - *Part of HTTP, URI, and HTML family*
 - *Request-response protocol*
- One socket for single request-response
 - *original specification*
 - *have changed due to performance issues*
 - *many concurrent requests*
 - *overhead when establishing same connections*
 - *HTTP 1.1 offers persistent connection and pipelining*
 - *Domain sharding*
- HTTP is stateless
 - *Multiple HTTP requests cannot be normally related at the server*
 - *"problems" with state management*
 - *REST goes back to the original HTTP idea*

REST Core Principles

- REST architectural style defines constraints
 - *if you follow them, they help you to achieve a good design, interoperability and scalability.*
- Constraints
 - *Client/Server*
 - *Statelessness*
 - *Cacheability*
 - *Layered system*
 - *Uniform interface*
- Guiding principles
 - *Identification of resources*
 - *Representations of resources and self-descriptive messages*
 - *Hypermedia as the engine of application state (HATEOAS)*

HTTP Request and Response

- Request Syntax

```
method uri http-version <crLf>
(header : value <crLf>)*
<crLf>
[ data ]
```

- Response Syntax

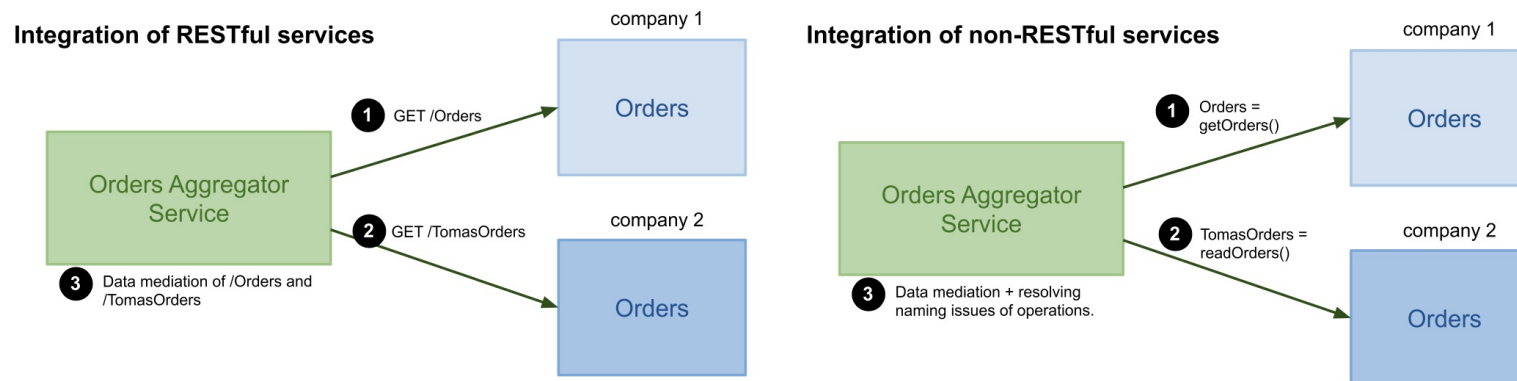
```
http-version response-code [ message ] <crLf>
(header : value <crLf>)*
<crLf>
[ data ]
```

- Semantics of terms

method	= "GET" "POST" "DELETE" "PUT" "HEAD" "OPTIONS"
uri	= [path] [";" params] ["?" query]
http-version	= "HTTP/1.0" "HTTP/1.1"
response-code	= valid response code
header : value	= valid HTTP header and its value
data	= resource state representation (hypertext)

Uniform Interface

- Uniform interface = finite set of operations
 - *Resource manipulation*
 - *CRUD* – Create (*POST/PUT*), Read (*GET*), Update (*PUT/PATCH*), Delete (*DELETE*)
 - *operations are not domain-specific*
 - For example, **GET /orders** and not **getOrders()**
 - This reduces complexity when solving interoperability
- Integration issues examples



Safe and Unsafe Operations

- Safe operations
 - *Do not change the resource state*
 - *Usually "read-only" or "lookup" operation*
 - *Clients can cache the results and refresh the cache freely*
- Unsafe operations
 - *May change the state of the resource*
 - *Transactions such as buy a ticket, post a message*
 - *Unsafe does not mean dangerous!*
- Unsafe interactions and transaction results
 - **POST** response may include transaction results
 - *you buy a ticket and submit a purchase data*
 - *you get transaction results*
 - *and you cannot bookmark this..., why?*
 - *Should be referable with a persistent URI*

Idempotence

- Idempotent operation
 - *Invoking a method on the same resource always has the same effect*
 - *Operations GET, PUT, DELETE*
- Non-idempotent operation
 - *Invoking a method on the same resource may have different effects*
 - *Operation POST*
- Effect = a state change

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Persistent connections

- Persistent HTTP connection = HTTP keepalive
 - *TCP established connection used for multiple requests/responses*
 - *Avoids TCP three-way handshake to be performed on every request*
 - *Reduces latency*
 - *FIFO queuing order on the client (request queuing)*
 - *dispatch first request, get response, dispatch next request*
- Example: **GET /html**, **GET /css**
 - *server processing time 40ms and 20ms respectively*
- Without HTTP keepalive
 - *three-way handshake 84ms before the data is received on the server*
 - *Response received at 152ms and 132ms respectively*
 - *The total time is 284ms*
- HTTP keepalive
 - *One TCP connection for both requests*
 - *In our example this will save one RTT, i.e. 56ms*
 - *The total time will be 228ms*

Persistent connections savings

- Each request needs
 - *Without keepalive, 2 RTT of latency*
 - *With keepalive, the first request needs 2 RTT, a following request needs 1 RTT*
- Savings for **N** requests: **$(N-1) \times \text{RTT}$**
- Average value of **N** is 90 requests for a Web app
 - *Measured by HTTP Archive (<http://httparchive.org>) as of 2013*
 - *Average Web application is composed of 90 requests fetched from 15 hosts*
 - *HTML: 10 requests*
 - *Images: 55 requests*
 - *Javascript: 15 requests*
 - *CSS: 5 requests*
 - *Other: 5 requests*

HTTP pipelining

- Important optimization – response queuing
 - *Allows to relegate FIFO queue from the client to the server*
- Requests are pipelined one after another
 - *This allows the server to process requests immediately one after another*
 - *This saves one request and response propagation latency*
 - *In our example, the total time will be 172ms*
- Parallel processing of requests
 - *In our example this saves another 20ms of latency*
 - **Head of line blocking**
 - *Slower response (css with processing time 20ms) must be buffered until the first response is generated and sent (no interleaving of responses)*
- Issues
 - *A single slow response blocks all requests behind it*
 - *Buffered (large or many) responses may exhaust server resources*
 - *A failed response may terminate TCP connection*
 - *A client must request all sub-sequent resources again (duplicate processing)*
 - *Some intermediaries may not support pipelining and abort connection*
- HTTP pipelining support today is limited

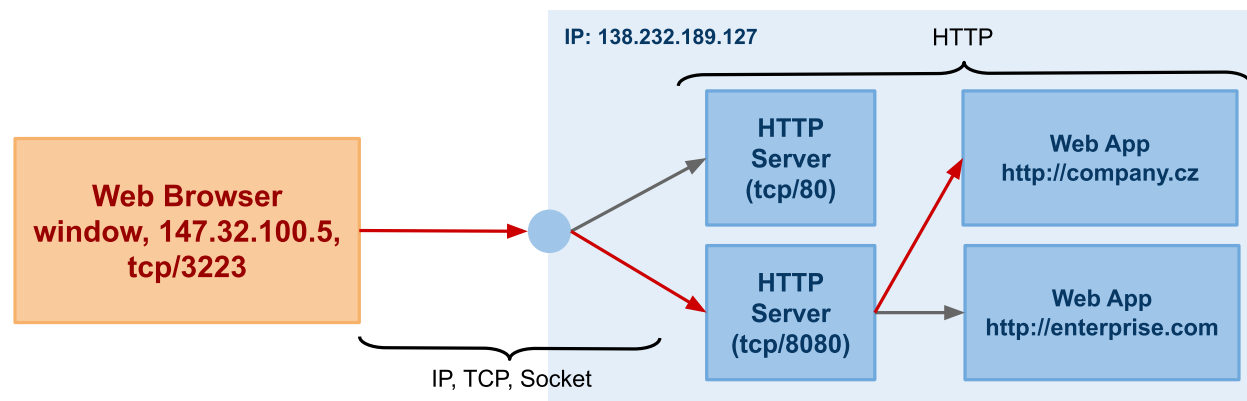
Multiple TCP connections

- Using only one TCP connection is slow
 - *Client must queue HTTP requests and process one after another*
- Multiple TCP connections work in parallel
- **There are 6 connections per host**
 - *The client can dispatch up to 6 requests in parallel*
 - *The server can process up to 6 requests in parallel*
 - *This is a trade-off between higher request parallelism and the client and server overhead*
- The maximum number of connections prevents from DoS attacks
 - *The client could exhaust server resources*
- Domain sharding
 - *The connection limit as per host (origin)*
 - *There can be multiple origins used in a page*
 - *Each origin has 6 maximum connection limit*
 - *A domain can be sharded*
 - **www.example.com** → **shard1.example.com, shard2.example.com**
 - *Each shard can resolve to the same IP or different IP, it does not matter*
 - *How many shards?*

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Serving HTTP Request



- Serving HTTP request

1. User enters URL **http://shard1.example.com/orders** to the browser
2. DNS resolution: browser gets an IP address for **shard1.example.com**
3. Three-way handshake: browser and Web Server creates a socket
4. Browser sends ACK and HTTP request:
 - 1 | GET /orders HTTP/1.1
 - 2 | Host: shard1.example.com
5. Web server passes the request to the web application **shard1.example.com** which serves **GET orders** and that writes a response back to the socket.

Virtual Host

- Virtual host
 - *Configuration of a named virtual host in a Web server*
 - *Web server uses host request header to distinguish among multiple virtual hosts on a single physical host.*
- Apache virtual host configuration
 - *Two virtual hosts in a single Web server*

```
1  # all IP addresses will be used for named virtual hosts
2  NameVirtualHost *:80
3
4  <VirtualHost *:80>
5      ServerName www.example.com
6      ServerAlias shard1.example.com shard2.example.com
7      ServerAdmin admin@example.com
8      DocumentRoot /var/www/apache/example.com
9  </VirtualHost>
10
11 <VirtualHost *:80>
12     ServerName company.cz
13     ServerAdmin admin@firm.cz
14     DocumentRoot /var/www/apache/company.cz
15 </VirtualHost>
```

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HTTP in Kubernetes

- Kubernetes does not provide built-in Layer 7 (HTTP) routing
- Services expose Pods at TCP/UDP level
 - **ClusterIP**: *internal-only*
 - **NodePort**: *accessible via NodeIP:NodePort*
 - **LoadBalancer**: *cloud-provider load balancer*
- For advanced routing (URLs, hostnames, TLS)
 - *Kubernetes uses the Ingress API*
- Ingress works at the HTTP layer (L7) and provides:
 - *Path-based routing*
 - *Host-based routing*
 - *TLS termination*
 - *Centralized access point*

Ingress Resource

- An Ingress is a set of HTTP routing rules
- It defines how external HTTP/S traffic reaches cluster Services
- An Ingress needs an Ingress Controller
 - *NGINX Ingress Controller*
 - *HAProxy Ingress*
 - *Traefik*
 - *Contour*
- Controller watches Ingress objects and configures
 - *reverse proxy*
 - *load balancer*
 - *dynamic routing tables*

How Ingress Works

- Client sends HTTP request to public IP
 - *LoadBalancer or NodePort*
- Ingress Controller receives the request
 - *Runs as a Deployment inside the cluster*
 - *Listens on ports 80/443*
 - *Uses reverse proxy (NGINX/HAProxy/etc.)*
- Ingress rules decide where to send the request
 - *Host matching* → **example.com**
 - *Path matching* → **/api, /app**
- The controller forwards the request to the Service
 - *The Service then forwards to Pods*
- Ingress Controller handles
 - *TLS termination*
 - *HTTP routing*
 - *Load balancing across Pods*

Ingress Example

- Example Ingress routing traffic to the **busybox-tcp-service**

```
1  apiVersion: networking.k8s.io/v1
2  kind: Ingress
3  metadata:
4    name: example-ingress
5    annotations:
6      nginx.ingress.kubernetes.io/rewrite-target: /
7  spec:
8    rules:
9      - host: busybox.example.com
10      http:
11        paths:
12          - path: /
13            pathType: Prefix
14            backend:
15              service:
16                name: busybox-tcp-service
17                port:
18                  number: 8080
```

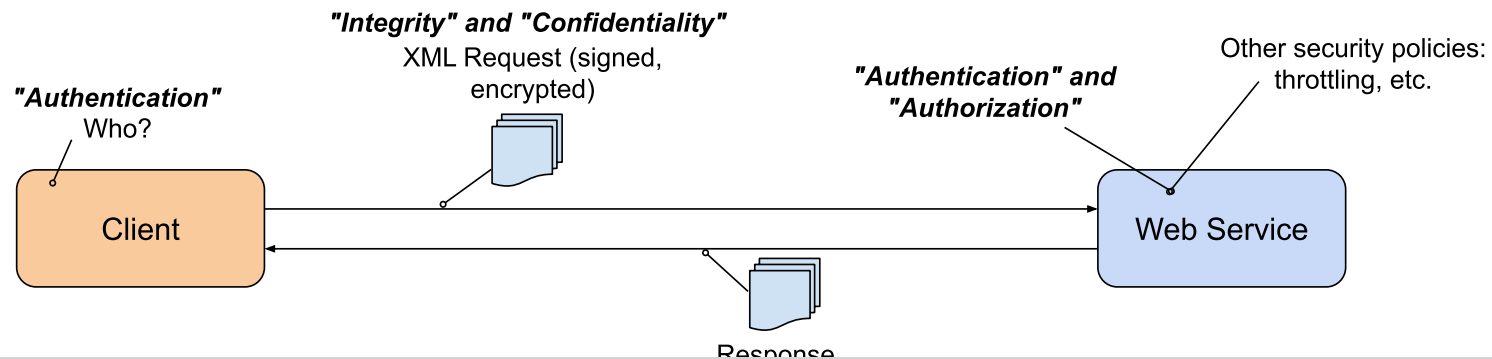
- Requests to **http://busybox.example.com/** are routed to the Service
- Ingress Controller performs HTTP reverse proxying and load balancing across Pods

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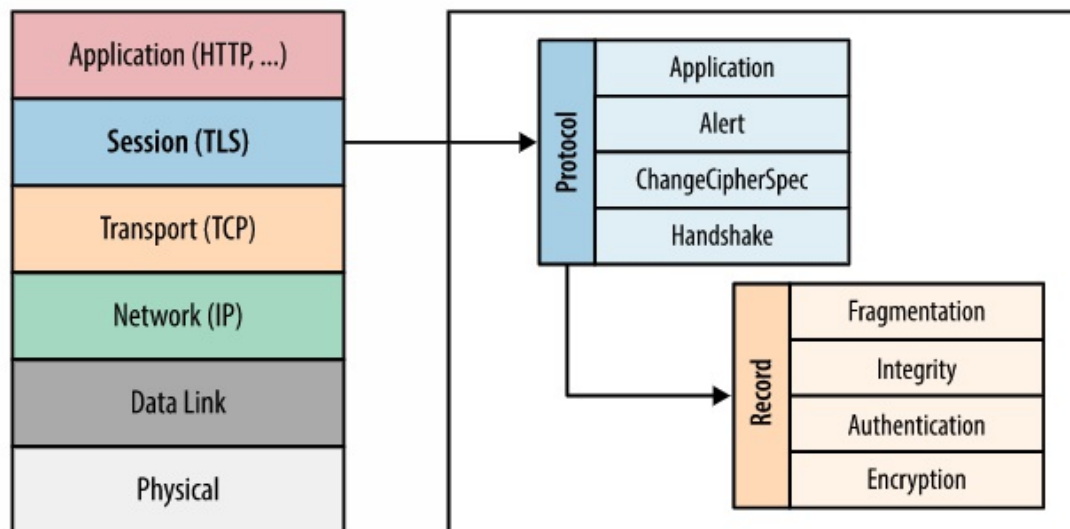
Web Service Security Concepts

- Securing the client-server communication
 - *Message-level security*
 - *Transport-level security*
- Ensure
 - *Authentication* – *verify a client's identity*
 - *Authorization* – *rights to access resources*
 - *Message Confidentiality* – *keep message content secret*
 - *Message Integrity* – *message content does not change during transmission*
 - *Non-repudiation* – *proof of integrity and origin of data*



Transport Level Security

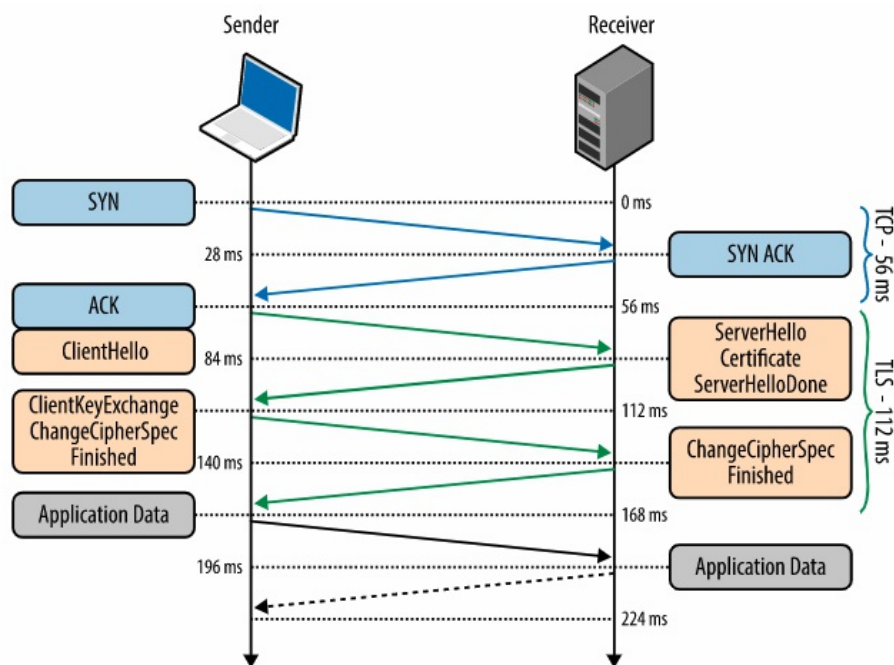
- SSL and TLS
 - *SSL and TLS is used interchangeably*
 - *SSL 3.0 developed by Netscape*
 - *IETF standardization of SSL 3.0 is TLS 1.0*
 - *TLS 1.0 is upgrade of SSL 3.0*
 - *Due to security flaws in TLS 1.0, TLS 1.1 and TLS 1.2 were created*
- TLS layer



TLS Services

- Encryption
 - *Peers must agree on ciphersuite and keys*
 - *This is achieved by **TLS handshake***
- Authentication
 - *Peers can authenticate their identity*
 - *The client can verify that the server is who it is claimed to be*
 - *Achieved by "Chain of Trust and Certificate Authorities"*
 - *The server can also verify the client*
- Integrity
 - *TLS provides message framing mechanism*
 - *Every message is signed with Message Authentication Code (MAC)*
 - *MAC hashes data in a message and combines the resulting hash with a key (negotiated during the TLS handshake)*
 - *The result is a message authentication code sent with the message*

TLS Handshake Protocol



- TLS Handshake

56 ms: ClientHello, TLS protocol version, list of ciphersuites, TLS options

84 ms: ServerHello, TLS protocol version, ciphersuite, certificate

112 ms: RSA or Diffie-Hellman key exchange

140 ms: Message integrity checks, sends encrypted "Finished" message

168 ms: Decrypts the message, app data can be sent

Key Exchange

- RSA key exchange(Rivest–Shamir–Adleman)
 - *The client generates a symmetric key*
 - *The client encrypts the key with the server's public key*
 - *The client sends the encrypted key to the server*
 - *The server uses its private key to decrypt the symmetric key*
- RSA critical weakness
 - *The same public-private key pair is used to:*
 - *authenticate the server (the server's private key is used to sign and verify the handshake)*
 - *encrypt the symmetric key*
 - *When an attacker gets hold of the server private key*
 - *It can decrypt the entire session*
- Diffie-Hellman key exchange
 - *Client and server can negotiate shared secret without its explicit communication*
 - *Attacker cannot get the key*
 - *Reduction of risk of compromising of the past communications*
 - *New key can be generated as part of every key exchange*
 - *Old keys can be discarded*

TLS and Proxy Servers

- TLS Offloading
 - *Inbound TLS connection, plain outbound connection*
 - *Proxy can inspect messages*
- TLS Bridging
 - *Inbound TLS connection, new outbound TLS connection*
 - *Proxy can inspect messages*
- End-to-End TLS (TLS pass-through)
 - *TLS connection is passed-through the proxy*
 - *Proxy cannot inspect messages*
- Load balancer
 - *Can use TLS offloading or TLS bridging*
 - *Can use TLS pass-through with help of Server Name Indication (SNI)*